Interior Point Methods for Linear Programming

Matteo De Francesco

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1 Introduction

In this report we are going to review the basics of the interior point methods for linear programming problems with two different approaches: logarithmic barrier methods and primal-dual interior point methods, both using the Newton approach to find the directions of improvements.
Let's consider a standard linear optimization problem

$$\begin{cases} \min_x & c^\intercal x \\ s.t. & Ax = b \\ & x \geq 0 \end{cases} \tag{P}$$

and its analogous dual problem

$$\begin{cases} \max_{y,s} & b^\intercal y \\ s.t. & A^\intercal y + s = c \\ & s \geq 0 \end{cases} \tag{D}$$

with $x, c, s \in \mathbb{R}^n$, $b, y \in \mathbb{R}^m$, $A \in \mathbb{R}^{m \times n}$.
The duality gap between the (P) and (D) can be computed as

$$c^\intercal x - b^\intercal y = x^\intercal s \geq 0$$

2 Newton's Method for Primal Logarithmic Barrier Optimization Problem

In the barrier method, we include the constraints $x \geq 0$ into the objective function by introducing a smart way of representing them through the logarithm function. In this way we penalize the objective function towards $+\infty$ if constraints are violated.
We can rewrite then the primal problem in the following way

$$\begin{cases} \min_x & c^\intercal x - \mu \sum_{j=1}^n \ln x_j \\ s.t. & Ax = b \end{cases} \tag{$P(\mu)$}$$

having the objective function

$$f_\mu(x) = c^\intercal x - \mu \sum_{j=1}^n \ln x_j$$

Since we apply the Newton method, we immediately compute the gradient and the Hessian of the above objective function. Given a feasible $\bar{x}$

$$\nabla f_\mu(\bar{x}) = c - \mu \bar{X}^{-1} e$$

and

$$\nabla^2 f_\mu(\bar{x}) = \mu \bar{X}^{-2}$$

where:

$$\bar{X} = \begin{bmatrix} x_1 & 0 & \cdots & \cdots & 0 \\ 0 & x_2 & \cdots & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & \cdots & \cdots & x_n \end{bmatrix} \qquad\qquad e = \begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix}^\intercal$$

and so $\bar{X}^{-1}$ and $\bar{X}^{-2}$ are respectively the element-wise inverse and the element-wise inverse squared.

In general Newton step is the solution of the second-order quadratic approximation of an optimization problem at the current point, which we chosen to be $x = \bar{x}$. We can write down the second-order approximation of $f_\mu(x)$ at $x = \bar{x}$

$$\begin{aligned} \min_x \quad & f_\mu(\bar{x}) + \nabla f_\mu(\bar{x})^\intercal (x - \bar{x}) + \tfrac{1}{2}(x - \bar{x})^\intercal \nabla^2 f_\mu(\bar{x})(x - \bar{x}) \\ s.t. \quad & A(x - \bar{x}) = 0 \end{aligned}$$

Rewriting $(x - \bar{x}) = \Delta x$ and substituting the definition of gradient and hessian inside the objective function we can rewrite everything as

$$\begin{aligned} \min_{\Delta x} \quad & (c - \mu\bar{X}^{-1}e)^\intercal \Delta x + \tfrac{1}{2}(\Delta x)^\intercal (\mu\bar{X}^{-2})(\Delta x) \\ s.t. \quad & A\Delta x = 0 \end{aligned}$$

We can write then the KKT conditions for this optimization problem:

$$\begin{cases} c - \mu\bar{X}^{-1}e + \mu\bar{X}^{-2}\Delta x = A^\intercal y' \\ A\Delta x = 0 \end{cases}$$

which by manipulation we get the Newton directions for variables x and y

$$\begin{cases} \Delta x = \bar{X}\left[I - \bar{X}A^\intercal (A\bar{X}^2A^\intercal)^{-1}A\bar{X}\right](e - \frac{1}{\mu}\bar{X}c) \\ y' = (A\bar{X}^2A^\intercal)^{-1}A\bar{X}(\bar{X}c - \mu e) \end{cases}$$

Now, we can easily update our variables by the following rules:

$$x' = x + \Delta x$$
$$y = y'$$
$$s' = c - A^\intercal y'$$

More practically, we will solve directly the following system of equations

$$\begin{bmatrix} \mu\bar{X}^{-2} & -A^\intercal \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ y' \end{bmatrix} = \begin{bmatrix} \mu\bar{X}^{-1}e - c \\ 0 \end{bmatrix}$$

3 Primal-Dual Path following IPM

In the previous solution primal and dual variables are not treated in the same way. Instead primal-dual Interior Point methods follow the same treatment.

Let $(\bar{x}, \bar{y}, \bar{s})$ be our current iterate, which we assume is primal and dual feasible and for which are inequalities are strict, namely

$$A\bar{x} = b, \bar{x} > 0 \qquad\qquad A^{\intercal}\bar{y} + \bar{s} = c, \bar{s} > 0$$

Let us introduce a direction of change $(\Delta x, \Delta y, \Delta s)$, next iterate will be given by $(\bar{x}, \bar{y}, \bar{s}) + (\Delta x, \Delta y, \Delta s)$, and we would like this iterate to solve the central path equations (i.e. the KKT system)

$$\begin{cases} A(\bar{x} + \Delta x) = b, \bar{x} + \Delta x > 0 \\ A^T(\bar{y} + \Delta y) + (\bar{s} + \Delta s) = c \\ \frac{1}{\mu}(\bar{X} + \Delta X)(\bar{S} + \Delta S)e - e = 0 \end{cases}$$

assuming that $(\bar{x}, \bar{y}, \bar{s})$ is primal-dual satisfies **??**

$$\begin{cases} A\Delta x = b - A\bar{x} \\ A^T\Delta y + \Delta s = c - A^{\intercal}\bar{y} - \bar{s} \\ \bar{S}\Delta x + \bar{X}\Delta s = \mu e - \bar{X}\bar{S}e \end{cases}$$

which can be solved efficiently exploiting a whatever linear system factorization is appropriate

$$\begin{bmatrix} A & 0 & 0 \\ 0 & A^{\intercal} & I \\ \bar{S} & 0 & \bar{X} \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta s \end{bmatrix} = \begin{bmatrix} b - A\bar{x} \\ c - A^{\intercal}\bar{y} - \bar{s} \\ \mu e - \bar{X}\bar{S}e \end{bmatrix}$$

So that we can update the variables with the following

$$\begin{aligned} x' &= x + \alpha_P \Delta x \\ y' &= y + \alpha_D \Delta y \\ s' &= s + \alpha_D \Delta s \end{aligned}$$

where the stepsizes α are modified as follows

$$\begin{aligned} \alpha_P &= \min\{1, r \min_{\Delta x_j < 0}\{\frac{\bar{x}_j}{-\Delta x_j}\}\} \\ \alpha_D &= \min\{1, r \min_{\Delta s_j < 0}\{\frac{\bar{s}_j}{-\Delta s_j}\}\} \end{aligned}$$

with r chosen equal to 0.99.

Regarding the stopping criterion we typically declare the problem to be solve when it is "almost" primal feasible, "almost" dual feasible, and there is "almost" no duality gap. Tolerance is set to $\varepsilon = 10^{-8}$, and we stop when:

$$\max\{\|A\bar{x} - b\|, \|A^{\intercal}\bar{y} + \bar{s} - c\|, \bar{s}^{\intercal}\bar{x}\} \leq \varepsilon$$